BRIDGING NEURAL AND BEHAVIORAL DATA WITH XAI IN PREDICTIVE NEUROMARKETING

25-26J-145-DS

Project Proposal Report

Ranawakaarachchi C P

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date

..... 31.08.2025

Mr. Samadhi Rathnayake

ABSTRACT

This research investigates how integrating neural signals (EEG data) and behavioral patterns (reaction times, gaze patterns) can improve consumer decision prediction in neuromarketing. Prior research has examined neural and behavioral data separately or through limited multimodal integration, yet their joint predictive value with interpretability remains underexplored.

This study will be conducted in **two phases**. In Phase One, synchronized publicly available EEG and behavioral datasets from decision-making tasks will be acquired and analyzed. In Phase Two, additional EEG data will be collected from volunteer participants under controlled experimental conditions to validate and extend the multimodal framework. All neural and behavioral signals will be preprocessed to remove noise and extract relevant features. Machine learning models will first be developed for unimodal analysis, followed by a multimodal deep learning model integrating EEG and behavioral features through attention-based fusion.

Expected outcomes include deeper insights into the interplay of brain activity and behavior during consumer decision-making, improved predictive accuracy through multimodal integration, and explainable outputs that enhance transparency and ethical alignment in neuromarketing models.

Keywords: EEG, Behavioral Data, Multimodal Learning, Consumer Insights, Neuromarketing, Explainable AI

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1. INTRODUCTION

In a modern market arena where consumers face a multitude of choices, decision making has become an increasingly complex process that is influenced by various cognitive, emotional, and social factors. In traditional approaches, consumer-based decision-making prediction has been based on gaze direction, response times, and self-reports. While these factors greatly contribute to the decision-making process, it often overlooks the nuances of the subconscious neural mechanisms that guides such behavior.

Therefore, Neuromarketing, an interdisciplinary field that bridges neuroscience, psychology, and data science aims to identify the processes behind this decision making, by tackling the core of the human brain and mind. Within this context, analyzing neural and behavioral data provides potential for producing models that are both predictive and interpretable. This research component focuses primarily on bridging neural data and consumer behavioral patterns thorough an integrated framework, and aims to advance consumer insights through multimodal learning techniques.

1.1. Background and Literature Survey

Electroencephalography (EEG) is a test that detects electrical activities in the human brain. The electrical signals received through electrodes are amplified and changes are mapped out on a graph. EEG, when used alongside behavioral data offers a complementary perspective of brain activity, which will allow researchers to study the correlation between them. This will provide a more comprehensive understanding on consumer decision-making based on these factors. This dual-perspective integration will be critical for building predictive and interpretable models in neuromarketing.

Existing research in neuromarketing has shown that decision biases that cannot be revealed through behavioral data, can be extracted through the analysis of neural signals.

Tanveer et al. (2024) demonstrated that multimodal fusion of EEG with eye-tracking signals can improve consumer choice prediction. [1] Their approach while highlighting the advantage of integrating behavioral data with neural signals, has limitations regarding interpretability, as primarily only focused on the predictive capabilities, rather than the explainability of the results. Bryne et al. (2022) further emphasized how most neuromarketing research, while utilizing EEG-based models, still uses unimodal approaches rather than integrated multimodal pipelines [2].

Zhang and Lin (2021) similarly reviewed studies combining EEG with eye-tracking [3]. They concluded with the analysis that neural and behavioral metrics can improve prediction performance.

Usman et al. (2025) also found significantly improved predicted accuracy while investigating consumer choice prediction using EEG and eye-tracking, compared to other unimodal baselines [4]. As in earlier studies, interpretability has not been brought a main focus. Further emphasis on the usage of advanced multimodal learning techniques to capture the cross-modal relationships was given in Wu et al. (2020) in which deep correlation analysis (DCCA) to integrate EEG functional connectivity with eye movement data for emotion recognition yielded classification accuracies of above 85% [5].

Ishtiaque et al. (2025) utilized EEG metrics such as beta/alpha engagement indices to predict viewers' preference for socially aware advertisements [6]. In this study, consumer-based prediction was based on neural data alone, and behavioral context as largely absent. The study Panteli et al. (2024) further emphasized that EEG measures such as alpha asymmetry and late positive potentials, when combined with behavioral signals such as eye-tracking and facial expression analysis, are reliable predictors of consumer responses [7].

Finally, Kondakar et al. (2024) noted the increasing shift towards hybrid models in neuromarketing research, which combine behavioral interaction with EEG data [8] consisting of multimodal fusion, interpretability, and ethical safeguards. These standards seem to emerge as important trends moving forward. Collectively, these studies while integrating EEG with behavioral metrics, show a lack of advanced fusion multimodal architecture and explainable AI methods leaving a gap that this research component aims to address.

1.2. Research Gap

Although EEG and behavioral measures have been individually studied for predicting consumer decision, most approached prioritize unimodal models to emphasize on one aspect of the two, leaving their joint predictive power underexplored. Recent multimodal studies that demonstrate improved accuracy through fusion lacks interpretability and adaptability. Therefore, this leaves a gap for integrated frameworks that while combining neural and behavioral signals also provide interpretable, adaptive, and ethically aligned consumer predictions.

1.3. Research Problem

How can neural signals (EEG) and behavioral data (reaction times, gaze patterns) be effectively integrated into a multimodal framework that improves prediction of consumer preferences, by discovering the correlation between neural and behavioral responses, while ensuring interpretability, adaptability, ethical transparency, and validation through real-world EEG data collected from volunteer participants?

2. OBJECTIVES

2.1. Main Objective

Develop a multimodal framework that integrated EEG based neural signals with behavioral metrics to explore their joint prediction power in predicting consumer preferences and decisons with improved accuracy interpretability and transparency

2.2. Sub Objectives

- Collect and preprocess synchronized EEG and behavioral data from publicly available decision-making datasets.
- Design and conduct a controlled experiment to collect EEG data from volunteer participants for validation purposes.
- Analyze the individual predictive performance of EEG-only and behavior-only machine learning models.
- Design and implement a multimodal attention-based fusion model integrating both EEG and behavioral features.
- Evaluate predictive performance (accuracy, F1, AUC) across unimodal and multimodal approaches, including validation using volunteer-collected EEG data.
- Provide interpretable consumer insights by analyzing how neural and behavioral data jointly influence choice behavior.

3. METHODOLOGY

3.1 . System Architecture

1. **Data Acquisition** – EEG datasets will be obtained from repositories such as OpenNeuro and DEAP, while behavioral datasets (eye-tracking, reaction times) will be obtained from repositories such as MAMEM. In addition, EEG signals will also be collected from a small number of volunteer participants during controlled decision-making tasks to support the analysis.
2. **Preprocessing** –
   a. **EEG:** Bandpass filtering, artifact removal (ICA), frequency-band extraction (alpha, beta, theta).
   b. **Behavioral:** Reaction time normalization, gaze fixation clustering, saccade metrics.
3. **Feature Engineering** – Convert EEG signals into time-frequency representations and encode behavioral patterns into temporal sequences.
4. **Unimodal Baselines** – CNN-LSTM models will be used for EEG data, while LSTM/GRU models will be used for behavioral sequences.
5. **Multimodal Fusion** – An attention-based fusion network will be implemented to integrate EEG and behavioral features at the representation level.
6. **Evaluation** – Compare unimodal and multimodal models using evaluation metrics such as accuracy, F1-score, and AUC.
7. **Explainability** – Apply SHAP values to analyze the importance of neural and behavioral features and generate interpretable insights into consumer decision-making.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

- Import and preprocess EEG and behavioral datasets from public repositories.
- Conduct controlled EEG data collection from volunteer participants.
- Build baseline predictive models for each modality.
- Implement attention-based multimodal fusion.
- Validate model using real-world EEG data.
- Generate explainable outputs via SHAP and counterfactual analysis.

- Provide interpretable visualizations of neural-behavioral interaction in consumer decisions.

4.2. System Requirements

System Overview Diagram

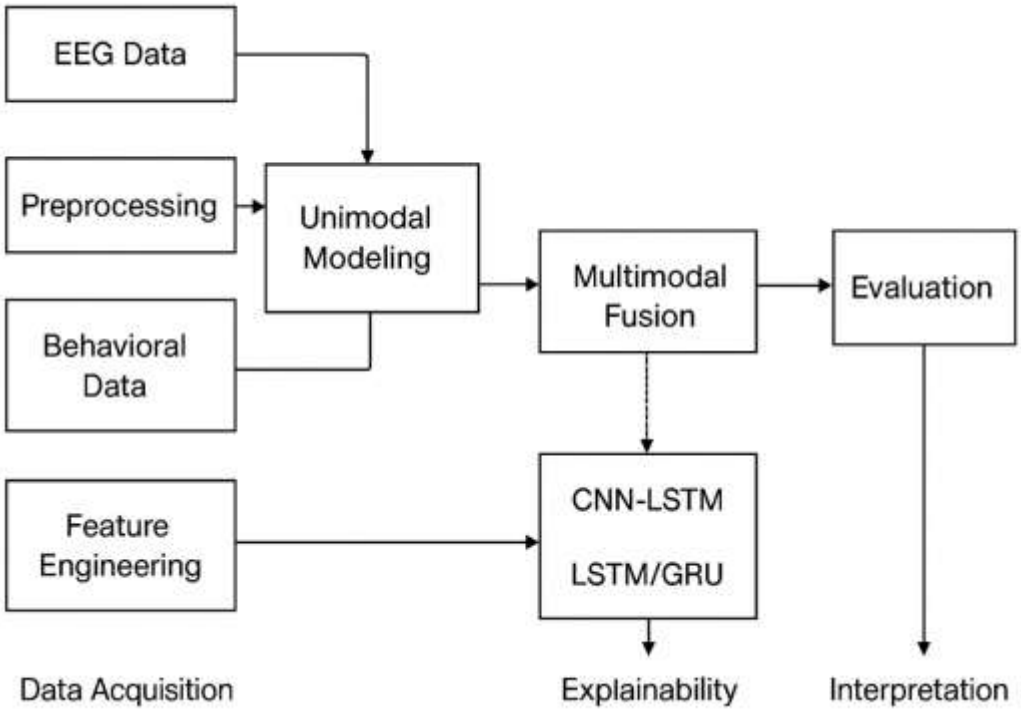

Figure 01: System Overview Diagram

- **Hardware**: GPU-enabled workstation (NVIDIA CUDA support).
- **Software**: Python, PyTorch/TensorFlow, SHAP, MNE.
- **Datasets**: EEG (OpenNeuro, DEAP) + behavioral (MAMEM).

4.3. User Requirements

- Researchers should be able to analyze EEG and behavioral interactions transparently.
- The system must support validation using real-world EEG recordings.
- Ethical compliance must be maintained for volunteer data collection.
- End-users (e.g., marketers) should gain interpretable insights rather than opaque black-box predictions.
- Outputs must emphasize ethical transparency and responsible use of neural consumer data.

5. PERSONNEL AND FACILITIES

• **Researcher:** Ranawakaarachchi C. P. (IT22083746) – Responsible for data preprocessing, modeling, evaluation, and experimental coordination.
• **Supervisor Support:** Guidance on neuroscience interpretation, experimental design, and machine learning methodology.
• **External Supervisor:** Oversight on EEG acquisition and ethical compliance.
• **Facilities:** Access to high-performance GPU computing (SLIIT labs), EEG recording equipment (clinical/research EEG unit), Controlled experimental environment for volunteer recordings, Dataset repositories (OpenNeuro, MAMEM).

6. GANTT CHART

Table 01

7. WORK AND BREAKDOWN STRUCTURE

1. Data Handling

• Collect public EEG & behavioral datasets
• Design volunteer experiment protocol
• Conduct EEG recordings
• Preprocess signals and extract features

2. Modeling

• Build unimodal baselines (EEG, behavioral)
• Implement multimodal fusion
• Fine-tune with collected EEG data

3. Evaluation

• Compare unimodal vs multimodal performance
• Validate with volunteer EEG
• Cross-phase robustness analysis

4. Explainability

• Apply SHAP and counterfactual analysis
• Analyze neural-behavioral contribution
• Ensure ethical transparency

5. Reporting

• Document methodology and findings
• Analyze experimental insights
• Prepare dissertation and presentation

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APPENDICES

Appendix A: Turnitin Plagiarism Checker

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